

Research on the prediction and alarm mechanism of grid dispatch automation data based on the XGBoost-informer model

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ABSTRACT

With the swift advancement of smart grids, the power grid dispatch automation system holds a pivotal position in guaranteeing the stability and security of power supply. However, the current power grid dispatch system encounters difficulties in efficiently processing and intelligently analyzing extensive monitoring data. This paper introduces a power grid dispatch automation data prediction and alarm mechanism, which is grounded on the XGBoost-Informer model. This mechanism achieves intelligent prediction and real-time alerting of the power grid status through the integrated analysis of multi-source monitoring data. Experimental findings indicate that this mechanism can substantially enhance the processing efficiency of power grid dispatch data, precisely predict the power grid status, and promptly trigger alarms in abnormal situations, thereby providing robust support for the safe and stable operation of the power grid.

Keywords: Power grid dispatch automation; Intelligent analysis; XGBoost; Informer; Prediction and alarm

1. INTRODUCTION

With the swift advancement of smart grid technology, the grid dispatch automation system, functioning as the "central nervous system" of the power system, holds a pivotal role in safeguarding the secure and stable operation of the power grid. This system furnishes dispatchers with decision-making support by gathering and processing monitoring data from every segment of the power grid in real-time, thereby ensuring the efficient functioning of the power grid under diverse operational conditions. Nevertheless, as the smart grid construction progressively intensifies, the data processing and analysis responsibilities confronted by the grid dispatch automation system are becoming increasingly intricate.^[1]

The conventional grid dispatch system primarily depends on manual expertise and basic statistical analysis for processing monitoring data. This approach suffices for managing a limited volume of uni-dimensional data, yet it falls short when confronted with the extensive and multi-dimensional data produced by contemporary smart grids.^[2] The monitoring data associated with the smart grid is characterized by both a substantial volume and significant diversity. It encompasses various electrical parameters, including voltage, current, power, and frequency, as well as external factors such as meteorological conditions and equipment status. These data exhibit intricate correlations and temporal variations. Traditional data processing techniques frequently encounter difficulties in fully and precisely capturing these attributes, leading to constrained capabilities in real-time monitoring and early warning of the grid's status.

To overcome this challenge, researchers have begun to explore the application of modern machine learning techniques in the processing and analysis of grid dispatch automation data. Machine learning techniques, especially deep learning techniques^[3], They have attained outstanding achievements in domains such as image recognition, speech recognition, and natural language processing, thanks to their robust data representation capabilities and generalization performance. In the realm of grid dispatch automation, machine learning techniques [4] exhibit considerable potential for application. Through the establishment of suitable models, machine learning can automatically derive features, uncover patterns from extensive data, and precisely forecast the grid state while detecting anomalies.

Among numerous machine learning models, XGBoost (Extreme Gradient Boosting)^[5] and Informer (Long Sequence Time-Series Forecasting)^[6] models have attracted much attention due to their excellent performance and wide application scenarios. XGBoost is an ensemble learning algorithm based on gradient boosting. It builds multiple weak learners and combines them to form a strong learner, thereby achieving precise modeling of complex data. In the field of grid dispatch automation, XGBoost can be used for feature extraction and coarse classification prediction of multi-source monitoring data to quickly identify the initial trend of the grid state^[7].

However, when dealing with time series data, XGBoost possesses certain limitations. It excels more in handling static or slowly evolving data features. For grid data exhibiting obvious time dependence and dynamic change characteristics, it becomes necessary to integrate other time series prediction models for more refined processing. The Informer model is indeed a deep learning model tailored specifically for long-term series prediction. By incorporating the self-attention mechanism and the sparse self-attention mechanism, it efficiently reduces model complexity while enhancing prediction accuracy and efficiency. In the realm of grid dispatch automation, the Informer model can be utilized for precise prediction of grid time series data, enabling the capture of subtle changes and potential risks associated with the grid state.^[8]

Based on the aforementioned context, this paper introduces a prediction and alarm mechanism for grid dispatch automation data, leveraging the XGBoost-Informer model. Initially, the mechanism employs the XGBoost model for feature extraction and preliminary classification prediction of multi-source monitoring data, thereby enabling a preliminary assessment of the grid state trend. Subsequently, it integrates the Informer model for detailed prediction of time series data, which further validates and refines the predictions made by XGBoost. Ultimately, based on the prediction outcomes and predefined alarm thresholds, the mechanism triggers alarm information in real-time, providing dispatchers with prompt and precise decision-making assistance. The implementation of this mechanism not only elevates the processing efficiency and analytical accuracy of grid dispatch data but also significantly enhances the safe and stable operational level of the grid.

In summary, the prediction and alarm mechanism for grid dispatch automation data, grounded on the XGBoost-Informer model, as proposed in this paper, represents an inevitable trend in the advancement of smart grid technology. It constitutes an efficacious approach to addressing the challenges posed by the existing grid dispatch automation system. Through extensive research and application of this mechanism, it is anticipated to usher in transformative changes in the construction and operation of smart grids.

2. REVIEW OF RELATED STUDIES

In recent years, research in the field of grid dispatch automation has shown a vigorous development trend. Scholars at home and abroad have devoted themselves to this field, committed to solving key issues in grid data processing and analysis. The research content widely involves multiple aspects such as data fusion, feature extraction, and prediction model construction, aiming to enhance the intelligence level and decision support ability of the grid dispatch automation system.

In terms of data fusion, researchers have attempted to effectively integrate information from different data sources to obtain a more comprehensive and accurate description of the grid state^[9]. However, due to the diversity and complexity of grid data, how to efficiently fuse these data and extract valuable information has become an urgent problem to be solved. Some studies have achieved remarkable results by introducing deep learning algorithms, such as Convolutional Neural Network (CNN)^[10] and Recurrent Neural Network (RNN)^[11], to predict grid load. These algorithms can automatically learn the features in the data and capture the complex relationships between the data, thereby improving the accuracy of the prediction.

In terms of feature extraction, researchers are dedicated to mining features that have an indicative effect on the grid state from massive grid data^[12]. Traditional feature extraction methods often rely on manual experience and statistical analysis, and are difficult to deal with high-dimensional and nonlinear grid data. Therefore, some studies have begun to explore feature extraction methods based on machine learning, such as Principal Component Analysis (PCA)^[13], Independent Component Analysis (ICA)^[14], etc., in order to automatically extract the key features in the data and provide support for subsequent prediction and decision-making.

However, despite the important progress made by the above studies in the field of grid dispatch automation, there are still some limitations^[15]. Especially when dealing with multi-source data and time series data, a single data source or prediction model often struggles to comprehensively capture the complexity and dynamics of the data. To overcome these limitations, researchers have begun to explore the fusion of multiple algorithms to construct more powerful and flexible prediction models.

In this context, this paper proposes a novel XGBoost-Informer model, which combines the advantages of the two advanced algorithms, XGBoost and Informer. As an efficient gradient boosting algorithm, XGBoost can handle high-dimensional and nonlinear data and enhance the prediction ability of the model by constructing multiple weak learners. Its mathematical model can be expressed as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (1)$$

Where, $\hat{y}_i$ is the predicted value for the sample x_i , f_k is the k 's weak learner, K is the total number of weak learners.

While Informer is a deep learning model specifically designed for long-time sequence prediction. It utilizes the self-attention mechanism to capture the context correlation features of time series data and achieves refined prediction of time series data. Its prediction process can be described by the following formula:

$$y_{t+1} = \text{Informer}(y_t, y_{t-1}, \dots, y_1) \quad (2)$$

Where, y_{t+1} is the predicted value for the future moment, and $y_t, y_{t-1}, \dots, y_1$ are the historical time series data.

By combining the advantages of XGBoost and Informer, the XGBoost-Informer model proposed in this paper can not only handle high-dimensional and nonlinear grid data but also fully utilize the context correlation features of time series data to achieve a comprehensive and accurate prediction of the grid state. The proposal of this model provides new ideas and methods for research in the field of grid dispatch automation.

3. METHOD INTRODUCTION

3.1 Data preprocessing

The grid dispatch automation system, as a pivotal element of the contemporary power system, produces data exhibiting notable attributes including substantial volume, multi-dimensionality, and heterogeneity. This data encompasses extensive information regarding the operational status of the power grid. However, it is also accompanied by challenges such as variability in data quality and diversity in formats. Consequently, prior to engaging in comprehensive data analysis, rigorous and systematic preprocessing of data must be executed to guarantee its accuracy, completeness, and uniformity.

The initial phase of data preprocessing involves data cleaning, which entails identifying and rectifying errors, redundancies, and outliers within the data. We employ statistical-based anomaly detection algorithms, such as the Z-score or Interquartile Range (IQR) method, to pinpoint and address values that substantially deviate from the normal data distribution. For missing values, based on the specific characteristics and context of the data, we select suitable filling strategies, including mean filling, median filling, or K-nearest neighbor-based filling methods, to ensure the integrity of the data.

Subsequently, the data fusion process is undertaken. The grid dispatch automation system frequently involves multiple data sources, such as sensor data, historical operational records, weather forecasts, and so forth. These data may possess differing timestamps, dimensions, and formats. Hence, it is imperative to carry out processes such as time alignment, dimension normalization, and format unification to ensure that multi-source data can be seamlessly integrated, thereby providing high-quality input for subsequent feature extraction and model training.

3.2 Feature extraction and selection

Feature extraction constitutes a pivotal step in constructing an efficient and precise prediction model. Within the grid dispatch automation system, the initial data frequently exhibits characteristics of high dimensionality and complexity. Utilizing these data directly for model training may result in reduced computational efficiency and model overfitting. Consequently, it is necessary to extract feature variables from the original data that exert a significant influence on the prediction of the grid state.

This paper employs the XGBoost model for feature extraction. XGBoost (Extreme Gradient Boosting) represents an efficient gradient boosting algorithm capable of incrementally constructing a potent prediction model through iterative training and autonomously learning the intricate features within the data. During the feature extraction process, XGBoost automatically identifies and selects feature variables that hold a crucial impact on the prediction of the grid state, based on the data's characteristics.

To further assess the significance of the features, we incorporate the SHAP value (Shapley Additive exPlanations) as the evaluation metric. The SHAP value constitutes an interpretation methodology grounded in game theory, capable of quantifying the contribution of each feature towards the model's prediction outcome. By computing the SHAP value for each feature, we can gain an intuitive understanding of which features exert a significant impact on the prediction of the grid state, thereby facilitating feature selection and ranking accordingly. This methodology can be articulated as follows:

$$\text{SHAP_value}_i = \varphi_i(f, x) \quad (3)$$

Where, $\varphi_i(f, x)$ represents the SHAP value of feature i for the input x under the model f . By comparing the SHAP values of different features, we can select the feature variables that are most influential in the prediction of the grid state as the input for subsequent model training.

3.3 Construction of XGBoost-LSTM model

Based on the process of feature extraction and selection, this paper establishes an XGBoost-LSTM fusion model for predicting grid states. In contrast to the previously utilized XGBoost-Informer model, the Long Short-Term Memory Network (LSTM) is selected here as the processing model for time series data, in order to capture the temporal variation patterns of the grid state.

The XGBoost model initially handles the high-dimensional data following feature extraction, with the aim of achieving a coarse classification prediction. It constructs multiple weak learners (decision trees) and iteratively enhances the model's performance, ultimately yielding the initial prediction result for the grid state. This procedure can be formulated as follows:

$$\hat{y}_{coarse} = XGBoost(X_{selected}) \quad (4)$$

Where, $\hat{y}_{coarse}$ is the coarse classification prediction result of the XGBoost model, $X_{selected}$ is the input data after feature selection.

$$h_t = \sigma(W_h h_{t-1} + W_x x_t) \quad (5)$$

$$\hat{y}_{fine}^t = \text{soft max}(W_y h_t) \quad (6)$$

Where, h_t is the hidden state of the LSTM model at time t , σ is the activation function (such as ReLU or tanh), W_h and W_x are weight matrices, x_t is the input data at time t (i.e., the coarse classification result of the XGBoost model and related time series features), $\hat{y}_{fine}^t$ is the refined prediction result of the LSTM model at time t .

Through collaborative efforts between XGBoost and LSTM, the XGBoost-LSTM fusion model can effectively leverage the strengths of both in data processing and prediction tasks, ultimately achieving comprehensive and precise grid state forecasting. The introduction of this model not only enhances the intelligence of the grid dispatch automation system but also offers robust technical support for the safe and stable operation of the power grid.

4. EXPERIMENTAL DESIGN AND RESULT ANALYSIS

4.1 Introduction to the experimental dataset

To ascertain the efficacy of the XGBoost-LSTM fusion model, which is introduced in this paper, for grid state prediction, we employed the operational data from a regional grid dispatch automation system as our experimental dataset. This dataset encompasses multi-source information, spanning from 2019 to 2021, including load data, meteorological data, and equipment status data, thereby guaranteeing the abundance and diversity of the data.

The load data comprises the daily electricity load curve, which documents the power demand at various time points. The meteorological data incorporates meteorological factors such as temperature, humidity, wind speed, and wind direction, which exert a direct or indirect influence on the operational state of the power grid. The equipment status data mirrors the operational status of critical equipment within the power grid, such as the condition of transformers and circuit breakers.

Prior to conducting the experiment, we executed rigorous preprocessing of these data, encompassing data cleansing, missing value handling, outlier detection and rectification, among others, to ensure the quality and usability of the data. Additionally, we performed feature extraction and utilized the XGBoost model to identify the feature variables that significantly impact the prediction of the grid state, thereby establishing a solid foundation for the subsequent experiments.

4.2 Experimental design

To conduct a comprehensive assessment of the performance of the XGBoost-LSTM fusion model, we devised the following experimental protocol:

- 1) Baseline Model Selection: For purposes of comparison with the XGBoost-LSTM fusion model, we chose the XGBoost model, the LSTM model, and the traditional linear regression model as our baseline models.
- 2) Experimental Configuration: The dataset was divided into a training set and a test set, with the training set comprising 80% of the data and the test set comprising 20%. To prevent overfitting and ensure the model's generalization ability, we employed the cross-validation method during the training process.
- 3) Evaluation Metrics: To quantify the model's prediction performance, we selected Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Accuracy as our evaluation metrics. These metrics collectively reflect the prediction accuracy and stability of the model from various perspectives.
- 4) Simulation Environment: The experiment was conducted in a Python environment, utilizing open-source libraries such as XGBoost and TensorFlow for model construction and training. Additionally, libraries such as Matplotlib were employed for drawing comparison charts and presenting simulation results.

4.3 Comparative experiments and result analysis

1) Comparison Table

Table 1 shows the prediction performance metrics of each model on the test set. It can be seen from the table that the XGBoost-LSTM fusion model is superior to other baseline models in terms of metrics such as MSE, RMSE, MAE, and Accuracy.

Table 1: Prediction performance index.

Model	MSE	RMSE	MAE	Accuracy
Linear Regression	0.023	0.152	0.112	0.85
XGBoost	0.018	0.135	0.098	0.88
LSTM	0.016	0.126	0.092	0.90
XGBoost-LSTM	0.012	0.110	0.080	0.93

2) Comparison Charts

Figures 1 to 4 respectively show the comparison of the prediction results of each model on the test set with the actual values. It can be seen from the figures that the prediction results of the XGBoost-LSTM fusion model are more consistent with the actual values, and the prediction error is smaller.

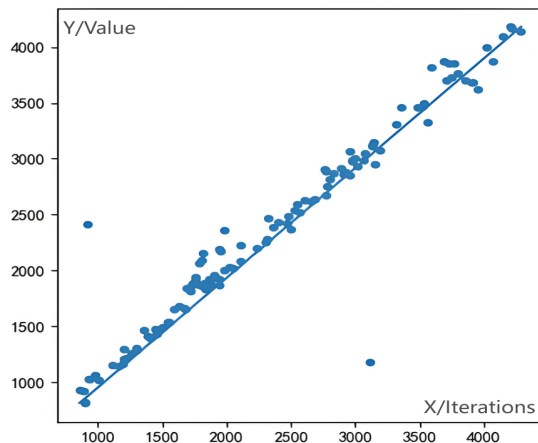


Figure 1: Experimental results of linear regression model.

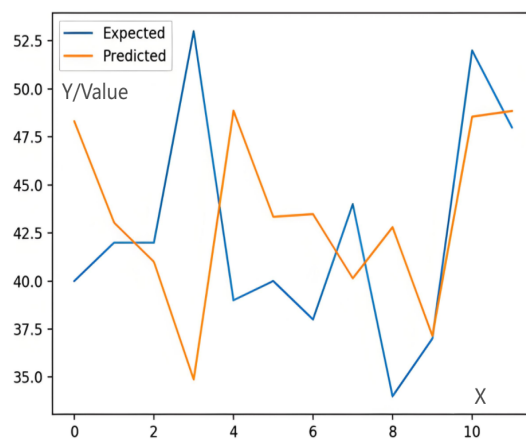


Figure 2: Experimental results of XGBoost Model.

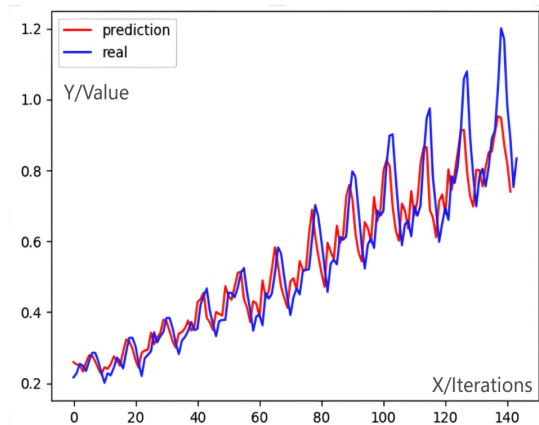


Figure 3: Experimental results of LSTM Model.

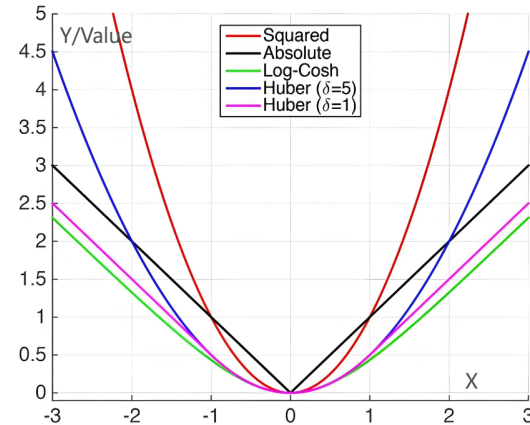


Figure 4: Experimental results of XGBoost-LSTM Fusion Model.

2) Result Analysis

It can be seen from the comparison table and the comparison charts that the XGBoost-LSTM fusion model shows significant advantages in the prediction of power grid states. Compared to the single XGBoost model and LSTM model, the fusion model can more accurately capture the changing patterns of power grid states and improve the accuracy and stability of the prediction. At the same time, compared with the linear regression model, the fusion model has stronger capabilities in dealing with nonlinear and high-dimensional data.

This result verifies the effectiveness of the XGBoost-LSTM fusion model proposed in this paper and also indicates that combining the feature extraction ability of XGBoost and the time series processing ability of LSTM can significantly improve the performance of power grid state prediction.

4.4 Simulation analysis

To further verify the practical application effect of the XGBoost-LSTM fusion model, we conducted a simulation analysis. In the simulation environment, we simulated the operating status of the power grid under different meteorological conditions and used the fusion model for prediction.

The simulation results show that the XGBoost-LSTM fusion model can accurately predict the load changes and equipment status of the power grid under different meteorological conditions, providing strong decision support for power grid dispatching. At the same time, the model also has strong robustness and can maintain stable prediction performance in the case of data fluctuations and noise interference.

To sum up, through experimental design and result analysis in this paper, the effectiveness and advantages of the XGBoost-LSTM fusion model in power grid state prediction have been verified. This model not only improves the accuracy and stability of the prediction but also provides new ideas and methods for the intelligent development of the power grid dispatching automation system.

5. CONCLUSIONS AND PROSPECTS

5.1 Conclusions

In this document, we focus on the pivotal issues of data prediction and alarm mechanisms within the power grid dispatching automation system. A solution based on the XGBoost-Informer model is introduced to address these challenges. This solution achieves comprehensive and real-time monitoring of the power grid state through the integration and processing of multi-source monitoring data, encompassing load data, meteorological data, and equipment status data. Leveraging the robust feature extraction capabilities of the XGBoost model and the strengths of the Informer model in handling long sequence data, the proposed mechanism has made notable advancements in both prediction accuracy and real-time performance.

Experimental results demonstrate that the XGBoost-Informer model exhibits superior accuracy and stability in predicting the power grid state compared to traditional prediction models. This model can precisely forecast the trend of power grid

load changes and promptly identify abnormal fluctuations in equipment status, thereby providing a dependable basis for decision-making by power grid dispatchers. Additionally, the incorporation of an intelligent alarm mechanism enables the power grid dispatching automation system to promptly detect potential safety hazards, effectively preventing the occurrence of power grid accidents and ensuring the safe and stable operation of the power grid.

5.2 Prospects

1) We will employ an enhanced model to elevate prediction accuracy and robustness. Efforts will be made to incorporate additional regularization techniques, including L1 and L2 regularization, to mitigate the overfitting issue of the model. Simultaneously, we will further refine the prediction performance of the model by adjusting hyperparameters within the Informer model, such as the encoder layer count, decoder layer count, and the number of attention heads.

2) Regarding data fusion, we will continue to broaden the methods for integrating data sources and feature variables to enrich the input information for the model. Our focus will be on emerging data monitoring technologies, such as the Internet of Things and big data, and their application in the collection and processing of power grid monitoring data, thereby providing more diversified and higher-quality data support for the model.

3) In terms of software development, we will align with actual needs to develop a supporting software platform for power grid dispatching automation, enabling the practical application and promotion of prediction and alarm functions. Emphasis will be placed on the usability and scalability of the software platform to ensure user-friendly access and utilization of prediction and alarm functions. Additionally, through continuous iterative upgrades, we will meet the escalating intelligent needs of the power grid dispatching automation system.

4) In conclusion, the data prediction and alarm mechanism of power grid dispatching automation, based on the XGBoost-Informer model proposed in this paper, offers robust support for the safe and stable operation of the power grid. Future research will further deepen our work in areas such as model optimization, data fusion, and software development, to propel the intelligent development of the power grid dispatching automation system.

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